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# Install BiocManager, which is the tool used to download specialized biology packages in
Rinstall.packages('BiocManager')

# Install Biostrings, a package used for handling and processing DNA and RNA sequences
BiocManager::install('Biostrings')

# Install the DECIPHER package, which is used for sequence alignment and building
phylogenetic trees BiocManager::install('DECIPHER')

# Load the Biostrings library into the current R session so its functions can be used
library(Biostrings)

# Load the DECIPHER library into the current R session so its functions can be used
library(DECIPHER)

# Create an empty container to hold the assembled sequence data from all six
samplesallcontigs <- c()

# Run a loop from 1 to 6 to automatically read the 'final.contigs.fa' file from each sample's
output folderfor(i in 1:6){

# Read the assembled DNA sequences for the current sample folder from the specified
computer directorycontigs <- readDNASTringSet(paste0('~/.Downloads/grace/t', i,
'_out/final.contigs.fa'))

# Combine the newly read sample sequences into our main collection list allcontigs <-
c(allcontigs, contigs) }

# Flatten the list structure so all of the sequences are combined into a single, uniform
dataset allcontigs <- do.call(c, allcontigs)

# Display the contents of the combined dataset to verify that all sequences loaded
correctly allcontigs

# Filter out small, incomplete genetic fragments and keep only the large sequences longer
than 5,000 base pairs toalign <- allcontigs[which(nchar(allcontigs) > 5000)]

# Rename the remaining six high-quality sequences simply as numbers 1 through 6
names(toalign) <- 1:length(toalign)

# Align the six filtered sequences side-by-side to find where their genetic letters match and
differ A <- AlignSeqs(toalign)

# Open a color-coded webpage in your browser to visually inspect the aligned genetic
sequences BrowseSeqs(A, htmlFile = "finalproject.html")

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# Create and display the final phylogenetic tree from the alignment using the Maximum  
Likelihood method tree <- Treeline(A, method = "ML", showPlot = TRUE)
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